§2 Formal definition of an ACS — part 3 of 5

`Reference: subsec:environment_space`

Source: `paper/section2_definition.tex`, lines 1579–2131 (§2.6), at commit `e6e12cd`. The Japanese manuscript is authoritative. Parts 1–2 are `02_definition_1.md` and `02_definition_2.md`.

2.6 The required environment space E and required-environment satisfaction

§2.6 fixes the semantics of the third component of the 7-tuple, the required environment space E. Its basic type was declared in `def:acs`(3) as $E \subseteq \{0,1\}^{X \times E_{\mathrm{ext}}}$, $E \neq \emptyset$. Here the constitutive requirement CR-1 is given, which constrains why each element of E is called "required". In addition, the set $E_{\mathrm{req}}(x)$ of external environment states in which all required factors are satisfied is defined as a derived object, and the subsection sets out the distinction between temporary non-satisfaction and lasting deficiency, the non-uniqueness of the factor decomposition, and the identifiability of necessity.

CR-1 is stated using the well-formed realization indicator $\mathrm{WF}_{i,t}$ (`def:wf_realization`) of §2.5. It does not use the collapse indicator $\mathrm{Col}_{i,t}$ (`def:collapse`), so that the semantic constraint on E does not depend on the definition of collapse.

New symbols: the required environmental factor η, the factor-wise satisfiable-state domain D_η, and the required-environment satisfaction set $E_{\mathrm{req}}(x)$ (the subscript "req" stands for "required"). $E_{\mathrm{req}}(x)$ is a subset of E_{ext} and is a different object from the required environment space E itself.

Required environmental factors

Definition (`def:environment_space`): required environmental factors and required environment space. Under `def:acs`(3), each element $\eta : X \times E_{\mathrm{ext}} \to \{0,1\}$ of E is called a **required environmental factor** of the ACS individual Σ_i, and E its **required environment space**. $\eta(x,e) = 1$ means that the factor is satisfied for the pair of internal state x and external environment state e, and $\eta(x,e) = 0$ that it is not. η takes the internal state x as an argument because, under the same external environment state, satisfaction can differ according to internal reserves, activity level, or the state of the self-side functions that use the required environment (`rem:env_utilization_function`). However, (CR-1c) of CR-1 excludes η being a condition on the internal state alone, independent of the external environment state. Since elements of E are elements of the set $\{0,1\}^{X \times E_{\mathrm{ext}}}$, two factors giving the same satisfaction judgment are identified. Phase 1 imposes no restriction on the cardinality of E.

A required factor is not a particular means of supply (`rem:factor_functional_class`). Elements of E are given not as particular substances, supply routes or devices, but as satisfaction judgments that bundle the routes that can satisfy them. For example, the availability of energy and nutrients needed for metabolism, or the possibility of continuous power supply, can each be represented as one η, while the individual means of supply that realize their satisfaction are not elements of E. This treatment absorbs substitutability inside η: if a particular means of supply were an element of E, that element would fail CR-1 when an alternative means exists. The granularity of functional bundling is implementation-dependent; see `rem:factor_nonuniqueness`.

Standard coarse classes of required factors (`rem:eta_coarse_classes`). As standard candidates for coarsely classifying required factors, the paper lists the set of four labels

$$\mathrm{Cls}_\eta := \{\mathrm{pow},\ \mathrm{mat},\ \mathrm{sink},\ \mathrm{cond}\} :$$

- pow (power / available free energy): necessity concerning the availability of the power or available free energy needed for the ACS's operation;
- mat (structural / material resources): necessity concerning the availability of material resources for maintaining, repairing and replacing structure;

- sink (dissipation / discharge): necessity concerning being able to dissipate or discharge heat, waste and other by-products to the outside;
- cond (external operating / boundary conditions): necessity concerning external conditions not reducible to pow, mat or sink — that the physical and environmental boundary states and constraints under which the ACS can operate, the possibility of receiving needed signals, access to objects, and the possibility of interaction are maintained.

This remark presents candidate classes; it is not a definition, constitutive requirement or assumption. The paper has not proved that the four classes are exhaustive or exclusive for every ACS. Hence: (a) classification by Cls_η is not a set-theoretic partition of E; (b) the same concrete external object can satisfy necessities under several labels; (c) what is classified is functional necessity, not concrete means of supply (`rem:factor_functional_class`); (d) it is not required that each ACS have factors corresponding to all four labels; (e) necessities arising in emergencies are not set up as an independent fifth label; (f) an emergency necessity that can be described by one of the four labels can be treated as a factor activated conditionally on the internal state x, which is possible because η takes x as an argument (`def:environment_space`); the paper does not claim that every emergency necessity can be described by the four labels; (g) finer classification is implementation-dependent and not fixed in Phase 1. Conceptually one may think of $\mathsf{Cls}_\eta \times \{\text{ordinary activation}, \text{conditional activation}\}$, but the paper does not introduce this product as a new formal type. How the ACS side can respond to changes in the external environment is a matter at a different theoretical level from the external dependence described by η, and is treated separately in §2.11.

Definition (`def:eta_satisfiable_domain`): factor-wise satisfiable-state domain (classification C). For each $\eta \in E$,

$$D_\eta := \{x \in X \mid \exists e \in E_{\text{ext}} :\ \eta(x, e) = 1\} \subseteq X$$

is called the **factor-wise satisfiable-state domain** of η. $x \in D_\eta$ means that, in internal state x, the factor can be satisfied under some external environment state; $x \notin D_\eta$ means that it is satisfied under no external environment state. D_η is a derived set fixed by η, X and E_{ext}, not an independent component of the 7-tuple; its defining formula does not use f. The symbol D_η is unrelated to the M1 output component $D_{M1,t}$ of §4. Its relation to the required-environment satisfiable-state domain D_{req} of §3 is stated in `def:req_domain`.

Required-environment utilization function (`rem:env_utilization_function`). With the factor η and E_{ext} fixed, whether an internal state x belongs to D_η does not depend on the realized external environment state e_t; it is fixed as a property of x. D_η itself, however, is relative to the choice of E_{ext}. The paper calls the involvement and cooperation of one or more existing functions in establishing the self-side conditions that place the internal state in D_η, collectively, the **required-environment utilization function** for that factor. The existing functions here can include basic functions corresponding to the primitive divisions (§2.9), the internal model structure and its update (§2.10, §2.11), and functions carried by the standard functional forms of TIPA, PSL and ISL (§4). For example, if the availability of breathable air is taken as one required factor, the cooperation of the self-side functions concerned with respiration and circulation can count as this. In relation to environmental utilization (EU) in §3, the required-environment utilization function is positioned as the collective name for the cooperation of existing functions (including the case of a single function) involved in the principle-response items of EU.

This remark gives a collective name; it introduces no new map, variable, 7-tuple component, mechanism, constitutive requirement or assumption. The required-environment utilization function is not treated as an independent group of functions alongside existing ones. It is recorded only as a candidate constitutively necessary function and is not given the classification-A label; examination of its constitutive necessity is a task for Phase 1.α (`subsec:phase1a`). Further:

- (a) If $x \notin D_\eta$ for some $\eta \in E$, then $E_{\text{req}}(x) = \emptyset$ (`def:E_req`): in an internal state where the self-side condition for that factor fails, required-environment satisfaction does not hold, whatever the external environment state.
- (b) Conversely, the paper does not state $E_{\text{req}}(x) \neq \emptyset$ as a general consequence of $x \in$

D_η for all $\eta \in E$, and assumes no implication from factor-wise satisfiability to joint satisfiability.

- (c) The paper does not state as a general consequence of $x \notin D_\eta$ by itself any of: $x \notin K$, blockage $A_t = \emptyset$ (`def:blockage_sec2`), $\mathrm{WF}_{i,t} = 0$ (`def:wf_realization`), functional termination (`def:functional_termination`) or collapse (`def:collapse`). Their relations depend on Π, f and CR-1.
- (d) The required-environment utilization function is distinguished from the environment-change response function of `rem:env_change_response`. The former concerns whether the self side is in a state able to use a given required environment; the latter concerns a cross-cutting function that can change present or future responses to changes in the external environment.

Constitutive requirement CR-1 (required-environment consistency)

`Reference: req:CR1`

For a 7-tuple structure Σ_i to be an ACS in the paper's sense, (CR-1a), (CR-1b) and (CR-1c) are required of its required environment space E.

(CR-1a) Necessity for persistence. A **state-transition sequence** is a sequence

$$(x_s, u_s, e_s)_{s\geq 0}, \qquad x_{s+1} = f(x_s, u_s, e_s), \quad u_s \in U, \quad e_s \in E_{\mathrm{ext}}$$

(`def:state_transition`). It is a mathematical sequence obeying the 7-tuple; its definition does not include as a requirement that well-formed realization holds at each time. Whether realization succeeds at each time is described separately by $\mathrm{WF}_{i,s}$. For each $\eta \in E$, the following are required.

- **(i) Realizability of deficiency.** There exist a state-transition sequence and a time $t_0 \geq 1$ such that
$$\mathrm{WF}_{i,t_0-1} = 1, \qquad \eta(x_{t_0-1}, e_{t_0-1}) = 1,$$
and $\eta(x_s, e_s) = 0$ for all $s \geq t_0$.
- **(ii) Inability to persist under deficiency.** For every pair of a state-transition sequence and a time t_0 satisfying the conditions of (i), there exists a finite time $T \geq t_0$ such that $\mathrm{WF}_{i,s} = 0$ for all $s \geq T$.

(CR-1b) Joint satisfiability.

$$\exists x \in X : \quad \{e \in E_{\mathrm{ext}} \mid \forall \eta \in E,\ \eta(x, e) = 1\} \neq \emptyset.$$

(CR-1c) External dependence (realizability of non-satisfaction due to the external environment state). For each $\eta \in E$, there exist a state-transition sequence and a time $t_0 \geq 1$ such that

$$\mathrm{WF}_{i,t_0-1} = 1, \qquad \eta(x_{t_0-1}, e_{t_0-1}) = 1, \qquad x_{t_0} \in D_\eta,$$

and $\eta(x_s, e_s) = 0$ for all $s \geq t_0$ (D_η is `def:eta_satisfiable_domain`).

Reading of the conditions.

- (CR-1a)(i) requires that lasting non-satisfaction of η be realizable as a state-transition sequence. Without it, a map with $\eta(x, e) = 1$ identically could be put in E; the antecedent of (ii) would never hold, and (ii) would hold vacuously. (i) excludes this and, together with (ii), requires that factors unnecessary for persistence not be in E. Imposing $\mathrm{WF}_{i,t_0-1} = 1$ and $\eta(x_{t_0-1}, e_{t_0-1}) = 1$ at $t_0 - 1$ identifies the start of non-satisfaction as t_0 and uses as evidence only sequences in which well-formed realization held just before the start. This makes it possible to take $T = t_0$ in (ii), so both the case where realization is lost at the start of non-satisfaction and the case where it is lost later can be represented.
- (CR-1a)(ii) requires that, when such non-satisfaction persists, realization from some finite time onward is not recovered.

- (CR-1b) excludes degenerate constructions in which the whole set of required factors cannot be satisfied jointly. It is a different condition from the factor-wise non-vacuity required by (CR-1a)(i), and the paper assumes no implication between them in either direction.
- (CR-1c) requires, for each factor, at least one state-transition sequence in which, with the internal state at the start of lasting non-satisfaction fixed, satisfaction and non-satisfaction of the factor are separated by the difference in the external environment state: at the start time t_0 the internal state belongs to D_η, so the factor could be satisfied under some external environment state, yet it is not satisfied under the realized e_{t_0}. This condition does not identify the external environment state as the cause of non-satisfaction, because the internal state can also change from $t_0 - 1$ to t_0. Without it, a map $\eta(x, e) = \rho(x)$ independent of the external environment state (for some $\rho : X \to \{0, 1\}$) would satisfy (CR-1a) as long as lasting non-satisfaction of ρ leads to loss of well-formed realization, and a condition on the internal state alone could be in E. For such η, $D_\eta = \{x \in X \mid \rho(x) = 1\}$, and $\eta(x_{t_0}, e_{t_0}) = 0$ means $x_{t_0} \notin D_\eta$, so (CR-1c) fails.
- In the evidence sequence of (CR-1c), $x_{t_0} \in D_\eta$ and $\eta(x_{t_0}, e_{t_0}) = 0$, so $\eta(x_{t_0}, e') = 1$ for some $e' \in E_{\text{ext}}$, and

$$\exists x \in X,\ \exists e, e' \in E_{\text{ext}} : \quad \eta(x, e) \neq \eta(x, e').$$

  That is, (CR-1c) implies that the satisfaction judgment depends on the external environment state. (CR-1c) also implies the existence requirement of (CR-1a)(i). (CR-1a)(i) is kept separately so that the range quantified in (CR-1a)(ii) is the whole of lasting non-satisfaction, including non-satisfaction starting in internal states outside D_η, not only non-satisfaction starting in D_η.
- (CR-1c) does not exclude η depending on the internal state (`def:environment_space`, `rem:env_utilization_function`). Nor does it quantify the degree of dependence on the external environment state or the range of internal states from which non-satisfaction in the above sense can start. How far to admit as required factors those whose external dependence appears only in some internal states, and how to quantify the degree, are tasks for Phase 2.

CR-1 is a condition placing the 7-tuple structure in the class of ACSs in the paper's sense; it is neither a diagnostic value at individual times nor an assumption restricting the range of later propositions (`rem:math_vs_realization`). Phase 1 gives no procedure for judging the satisfaction of CR-1 from finite observation; that formulation is a task for Phase 2 (`rem:necessity_identification`).

Relativity of (CR-1a)(i) to the external background (`rem:cr1_relativity`). Because (CR-1a)(i) requires that lasting non-satisfaction of η be realizable as a state-transition sequence, it is relative to the choice of E_{ext}. If E_{ext} contains no environment state with non-satisfaction of η, that η fails (i) and is not an element of E under that description. (CR-1c) is likewise relative to the choice of E_{ext}: if no state-transition sequence containing an external environment state that leaves the factor unsatisfied at a time when the internal state belongs to D_η can be realized, that η fails (CR-1c). This is consistent with the paper's framework, in which the 7-tuple is defined relative to E_{ext} (§2.2). How the identification of required factors depends on the choice of external background is a task for Phase 2.

How to read the quantification in (CR-1a)(ii) (`rem:cr1_quantification`). The antecedent of (CR-1a)(ii) states that non-satisfaction of η holds at every time from t_0 on. Since η takes both x_s and e_s as arguments, this antecedent excludes both of the following: first, $\eta(x_s, e_s) = 1$ being recovered through internal reserves or the like; second, $\eta(x_{s'}, e_{s'}) = 1$ being recovered because later $(x_{s'}, e_{s'})$ change through the selection of actions u_s. Recoverability through internal compensation and admissible actions is thus absorbed into the antecedent, and (CR-1a)(ii) concerns only non-satisfaction not recovered by these. The consequent states that $\mathrm{WF}_{i,s} = 0$ at every time from some finite time on. The value of T may differ by factor and by state-transition sequence. The condition gives no uniform upper bound on T, and does not exclude $T = t_0$, that is, realization being lost immediately at the start of non-satisfaction.

A necessary condition, not a sufficient one (`rem:cr1_one_way`). CR-1 states a one-directional implication from unrecovered non-satisfaction of a required factor to non-recovery of well-formed realization. The paper does not claim that satisfaction of the required factors suffices for the continuation of well-formed realization. In particular, the following implication is not asserted:

$$\forall \eta \in E:\ \eta(x_t, e_t) = 1 \implies \mathrm{WF}_{i,t+1} = 1.$$

Even when all required factors are satisfied, well-formed realization can be lost through damage to internal structure, the selection of actions, external actions or other causes. E is thus a component describing the side of necessary conditions for the persistence of the ACS; it does not give sufficient conditions for persistence.

Temporary non-satisfaction versus lasting deficiency (`rem:cr1_timing`). The paper does not state, from $\eta(x_t, e_t) = 0$ at some time by itself, either $\mathrm{WF}_{i,t} = 0$ or $\mathrm{WF}_{i,t+1} = 0$ as a general consequence. The antecedent of (CR-1a)(ii) requires persistence from t_0 on, which single or intermittent non-satisfaction does not meet. When non-satisfaction is resolved through internal reserves, recovery actions or changes in the external environment state, that state-transition sequence does not meet the antecedent of (CR-1a)(ii), and no consequence about the consequent arises. How much temporary non-satisfaction is tolerated, that is, the quantification of recovery margin, is not treated in Phase 1 and is a task for Phase 2 (L9 in `subsec:limitations_list`).

Required-environment satisfaction set

Definition (`def:E_req`): required-environment satisfaction set (classification C). For a state $x \in X$, the set of external environment states in which all required factors of E are satisfied is

$$E_{\mathrm{req}}(x) := \{e \in E_{\mathrm{ext}} \mid \forall \eta \in E,\ \eta(x, e) = 1\} \subseteq E_{\mathrm{ext}}.$$

With it, (CR-1b) can be written $\exists x \in X:\ E_{\mathrm{req}}(x) \neq \emptyset$. E_{req} is a correspondence from X to E_{ext}, a derived object fixed by E, X and E_{ext}. Its defining formula does not use f. However, the constitutive requirement CR-1 itself, which constrains why each element of E is called required, refers to state-transition sequences and the course of WF, and so depends on f. E_{req} is not an independent component of the 7-tuple.

Properties of E_{req} and distinction from other environment sets (`rem:E_req_properties`). As in `def:acs`(3), Phase 1 imposes no continuity or other regularity on η, so $E_{\mathrm{req}}(x)$ is in general neither a closed nor a compact subset of E_{ext}. For example, with $E_{\mathrm{ext}} = [0, 1]$ with the usual topology, $E = \{\eta\}$ and $\eta(x, e) = 1 \iff e < 1/2$, one has $E_{\mathrm{req}}(x) = [0, 1/2)$, which is neither closed nor compact (an example at the level of types). Propositions needing these properties state them as added assumptions where used. $E_{\mathrm{req}}(x)$ is fixed only by the satisfaction of required factors and does not state whether a feasible action exists under that environment state. Environment sets concerning one-step feasibility are defined separately in §2.8, and the two are distinguished. That is, neither the case where no feasible action exists under an external environment state in which all required factors are satisfied, nor the case where a one-step feasible action exists under an external environment state in which some required factor is not satisfied, is excluded.

Non-uniqueness of the factor decomposition (`rem:factor_nonuniqueness`). E need not be fixed uniquely down to the granularity of its decomposition into required factors. For example, forming the conjunction $\eta_\wedge(x, e) := \eta_1(x, e) \wedge \eta_2(x, e)$ of $\eta_1, \eta_2 \in E$ and setting

$$E' := (E \setminus \{\eta_1, \eta_2\}) \cup \{\eta_\wedge\},$$

E' gives the same required-environment satisfaction set $E_{\mathrm{req}}(x)$ as E, and may have the same status as E with respect to CR-1. The paper therefore does not require E to be unique down to the granularity of the factor decomposition. In Phase 1 one decomposition E is fixed for the ACS individual considered, and the existence of other coarse-grained or refined representations giving the same required-environment satisfaction set is not excluded. Examining additional conditions that would give uniqueness of the decomposition is a task for Phase 2.

Existence of necessity versus identifiability (`rem:necessity_identification`). The paper treats the required factors as fixed for the ACS individual independently of an observer's knowledge: that some η is required in the sense of CR-1 does not depend on whether it is observed or described. On the other hand, the paper gives no procedure for identifying E from finite observation for an unknown ACS individual. Observing responses under different external environment states, or predictions by an internal model, can be clues, but the paper presents no rule for recovering E from them. Empirical identification of required factors and the treatment of identification error are tasks for Phase 2. In this sense, descriptions concerning the existence of necessity are distinguished from descriptions concerning its identifiability.